Hydrodynamic Measurements of Jets in Crossflow for Gas Turbine Film Cooling Applications

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This paper presents the results of a detailed hydrodynamic study of a row of inclined jets issuing into a crossflow. Laser-Doppler anemometry was used to measure the vertical and streamwise components of velocity for three jet-to-mainstream velocity ratios: 0.25, 0.5, and 1.0. Mean velocity components and turbulent Reynolds normal and shear stress components were measured at locations in a vertical plane along the centerline of the jet from 1 diameter upstream to 30 diameters downstream of the jet. The results, which have application to film cooling, give a quantitative picture of the entire flow field, from the approaching flow upstream of the jet, through the interaction region of the jet and mainstream, to the relaxation region downstream where the flow field approaches that of a standard turbulent boundary layer. The data indicate the existence of a separation region in the hole from which the jet issues, causing high levels of turbulence and a relatively uniform mean velocity profile at the jet exit.

Introduction

A fundamental limitation to the development of improved film cooling for turbine blades is the lack of understanding of the fluid mechanisms affecting heat transfer during film cooling. Although there has been prolific research in the area of film cooling technology, much of this research has focused on the end product, i.e., the amount of heat transfer, while relatively few studies have attempted detailed descriptions of the flow field involved in the heat transport process. Comprehensive experimental measurements of jets-in-crossflow were made in this study to obtain a better understanding of the fundamental characteristics of this flow.

In previous studies of jets-in-crossflow, the most detailed measurements of the flow field have been made for a single jet issuing from a relatively long delivery pipe oriented normal to the mainstream. Although this is significantly different from the short length-to-diameter (L/D) inlet with the hole axis angled sharply downstream, generally used for film cooling, the flow field characteristics might be expected to be qualitatively the same. Crabb et al. (1981) used both laser-Doppler velocimetry (LDV) and X-sensor hot-wire anemometry to study a normal jet-in-crossflow with relatively high blowing ratios of $M = 1.15$ and 2.3, where M is the density–velocity product ratio of the jet to the mainstream. The jet issued from a pipe with $L/D = 30$. These investigators emphasized the importance of using LDV in the near field of the jet where the three-dimensionality, high turbulence, and reverse flow preclude the use of hot-wire techniques. Their measurements also indicated that the exit profile of the jet is highly distorted by the crossflow. For $M = 1.15$, measurements of the turbulent normal stresses showed a maximum at approximately 1/2 D downstream of the hole, which corresponded to a maximum in the mean streamwise velocity gradient located at $y/D = 2$.

The most detailed study of the turbulence characteristics of a jet-in-crossflow were made by Andreopoulos and Rodi (1984) for a normal jet issuing from a pipe with $L/D = 12$ and a smooth entrance. Using triple-sensor hot wires, measurements were made of all three velocity components. In this study, using a blowing ratio of $M = 0.5$, the crossflow was found to influence the pipe flow upstream of the exit, causing a highly nonuniform exit profile, which was skewed toward the back edge of the hole. The maximum turbulence level and uv shear stress levels occurred at locations from two to four diameters downstream of the jet and were coincident with maximum mean velocity gradients. The large velocity gradients in this region were due to a shear layer, which occurred on the lower edge of the jet because of a wake region below the jet.

Turbulence measurements for inclined jets have been limited to a few studies using single-sensor hot-wire probes. Since the accuracy of these probes is extremely poor in the highly turbulent, three-dimensional, and occasionally reversing flow field characteristic of jets-in-crossflow, these results should be regarded as qualitative descriptions of the flow turbulence. Kadotani and Goldstein (1979) and Yoshida and Goldstein (1984) used a row of jets introduced by pipes inclined at 35 deg to the surface and with $M = 0.35$ and $M = 0.50$, respectively. In both these studies mean and rms velocity measurements for the combined U and V components were presented for one streamwise position located one diameter downstream of the hole. At this point the maximum turbulence level was found to correspond with the lower edge of the injected flow where the mean velocity gradient was largest. Turbulence in-

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tensity was also measured by Jubran and Brown (1985) for holes inclined at 30 deg and with inlet pipe lengths of L/D = 15. Although two rows of holes were used, the second row was located x/D = 10 downstream of the first row, while the turbulence intensity measurements were made at x/D = 2.3 and 10. Results presented for blowing ratios of M = 0.2 and 1.0 were somewhat contradictory to the previous studies in that maximum turbulence levels were found at the point measured nearest to the wall (y/D = 0.1) rather than being coincident with the shear layer between the wake and the bottom edge of the jet.

In the present study, detailed measurements have been made of the mean and turbulence characteristics of jets-in-crossflow for hydrodynamic conditions typical of film-cooled turbine blades, but with a unity density ratio and a flat test surface. A short entrance of L/D = 3.5 was purposely used for the injected fluid to simulate actual film-cooled turbine blade configurations better, although this is significantly different from the pipe flow inlet configuration that has been used in past studies. An LDV was used to measure the distributions of streamwise and normal components of the mean and fluctuating velocity along with the uv shear stress over a range extending from one diameter upstream of the hole to 30 diameters downstream. Various sources of high turbulence were identified and the relaxation of the flow to essentially boundary layer characteristics was documented. A strong correspondence between the high turbulent normal and shear stresses with regions of high mean velocity gradients was also found.

Experimental Facility and Instrumentation

These experiments were carried out using a test plate installed in a recirculating low-speed wind tunnel. The test section, with dimensions 61 × 61 × 244 cm long, was preceded by a 6.25:1 contraction section with a bank of graduated screens and honeycomb upstream of the contraction. Air was used as the working fluid with the free-stream turbulence intensity in the test section measured at less than 0.5 percent at a velocitiy of 20 m/s. Air is injected into the mainstream through a row of 11 holes, 12.5 mm in diameter, in a flat plate. The holes are inclined at 35 deg and laterally spaced three diameters apart and are located 19 diameters downstream of the leading edge of the plate. Suction is used to remove the boundary layer at the test section inlet and a new boundary layer is initiated at the sharp leading edge of the plate. The length of the hole was made short relative to the diameter (L/D = 3.5) to simulate a typical film-cooled turbine blade geometry. Figure 1 shows a schematic of this arrangement. All 11 jets issue from a common plenum chamber located below the plate. Air to the plenum is supplied by a secondary flow loop containing a centrifugal fan and an orifice plate flowmeter. Different blowing ratios were obtained by varying the flow rate through the secondary flow loop while maintaining a constant mainstream velocity of 20 m/s.

Two-component velocity measurements were made with a TSI Inc. model 9100-10 LDV system. To allow measurements of the vertical velocity near the wall it was necessary to tilt the entire LDV 8 deg downward to prevent the lower beam from

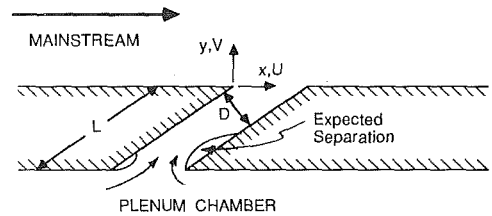

Fig. 1 Geometry of the 35 deg film cooling hole and coordinate system

intersecting the floor of the wind tunnel. As a result, the V component of velocity reported in this paper is composed primarily of the true vertical velocity, but also contains a small lateral velocity component. This tilt does not affect the measurement of the streamwise velocity component. The LDV was equipped with frequency shifting, which is necessary to distinguish reverse flow and also to avoid fringe bias errors in the measurement of the relatively small vertical velocity component. The diameter of the probe volume was reduced by a factor of 3.75 to d = 80 μm by passing the four transmitted beams through beam expansion optics. The receiving optics were modified to operate in a backscatter configuration offset 30 deg from the axis of the transmitted beams. As a result of this off-axis collection, the length of the probe volume was reduced by a factor of 5 to 1 = 160 μm. TSI counters were used for signal processing and an HP-1000 microcomputer system for data acquisition and anaysis. The HP-1000 system was also interfaced with the motion control system of the LDV traverse table to allow the computer to control the positioning of the table automatically. Velocity bias corrections, based on the residence time of a particle in the probe volume, were applied to the LDV data.

The mainstream and jets were independently seeded with 0.6 μm sugar particles generated by two TSI six-jet atomizers. To avoid bias due to different seed concentrations, the atomizers were adjusted to provide equal seed concentrations in the mainstream and jet flow. This condition was initially set and periodically checked by measuring the velocity and the data rate at various locations in the mainstream and in the jets. The seed concentration is proportional to the ratio of the data rate and the velocity. The jet air was obtained from the seeded mainstream, so the addition of a relatively small amount of seed was required to replace that which dropped out while passing through the secondary flow loop.

To obtain accurate measurements of the turbulence shear stress, essentially coincident measurements of the instantaneous streamwise and vertical components of velocity are necessary. A coincidence time window of 10 μs was used for these measurements, which allowed accurate measurements for flow fluctuations up to 50 kHz.

The time-averaged measurements presented in this paper were obtained using sample sizes of 1024 discrete coincident measurements. Measurement time for these samples varied from 10 s to 100 s. Precision uncertainties for these measurements were determined from repeatability tests and found to be ±0.5 percent for U and V, ±2 percent for u_{rms} and v_{rms}, ±15 percent for uv, where percentages are in terms of the free-stream velocity.

Nomenclature

D = hole diameter, Fig. 1
L = hole length, Fig. 1
M = density • velocity product ratio, jet-to-freestream
TL = turbulence level, two-dimensional
U = mean velocity, x direction
u_{rms} = rms velocity, x direction
uv = Reynolds uv shear stress
U_∞ = free-stream velocity
V = mean velocity, y direction
v_{rms} = rms velocity, y direction
x = streamwise distance, Fig. 1
y = distance from wall, Fig. 1
δ = boundary layer thickness, 99 percent velocity point
θ = boundary layer momentum thickness

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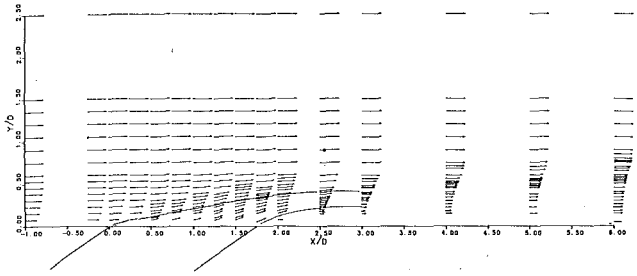

Fig. 2 Profiles of the normalized mean velocity vectors for the near-hole region of the jet exit and $M = 0.25$. Edges of the jet are defined by the interpolated mean streamlines.

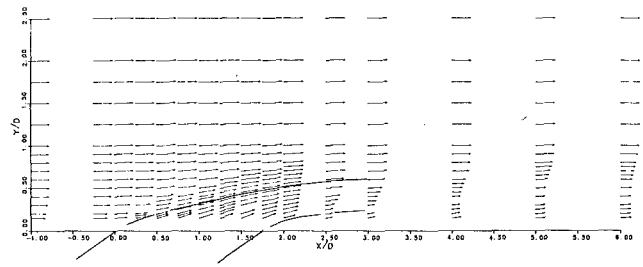

Fig. 3 Profiles of the normalized mean velocity vectors for the near-hole region of the jet exit and $M = 0.5$. Edges of the jet are defined by the interpolated mean streamlines.

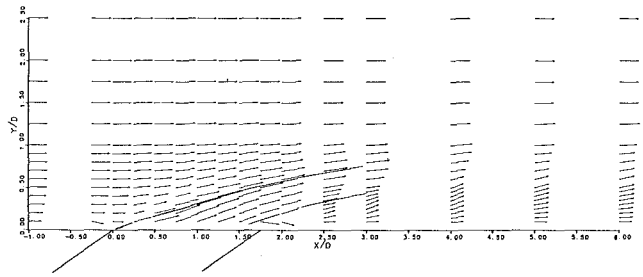

Fig. 4 Profiles of the normalized mean velocity vectors for the near-hole region of the jet exit and $M = 1.0$. Edges of the jet are defined by the interpolated mean streamlines.

Results and Discussion

Results are presented for two-component velocity measurements made along the centerline of the fourth jet and over a range from one diameter upstream of the jet to 30 diameters downstream. Three different blowing ratios were used with $M = 0.25$, 0.5, and 1.0, all with unity density ratio. These data will be discussed first in terms of the near-field mean flow and turbulence characteristics. Relaxation of the flow in the far field downstream of the hole will then be examined. Finally, results indicating the correspondence between mean velocity gradients and the turbulent normal and shear stresses will be presented.

For the following results the origin of the coordinate system is located on the wall at the leading edge of the jet. Because of the elliptical shape of the hole, the trailing edge is located at $x/D = 1.74$, where x is measured in the streamwise direction. The distance normal to the plate is denoted by y.

Mean Flow, Near Field. Profiles of mean velocity vectors for the three different blowing ratios are presented in Figs. 2, 3, and 4. The first two velocity profiles for each case were taken upstream of the hole at $x/D = -1.0$ and -0.25. The profile at $x/D = -1.0$ is the same for all three blowing ratios and indicates a boundary layer thickness of $\delta/D = 0.4$ ($\delta \approx 4.8$ mm). Only a slight deceleration is noticeable at $x/D = -0.25$, but for $M = 1.0$ there was a significant negative V component at the nearest wall location, which might indicate the existence of a streamwise vortex caused by the obstruction of the jet.

Measurements taken nearest the exit of the hole indicate relatively flat velocity profiles as the jet exists the hole. The magnitudes of the velocity vectors just past the jet exit are larger than might be expected for the respective blowing ratios because of an acceleration as the jets were diverted downstream by the crossflow. The velocity profile was relatively uniform for $M = 0.5$, but slightly skewed downstream for $M = 0.25$ and upstream for $M = 1.0$. This is in contrast to the findings of earlier studies, where highly nonuniform profiles were found using normal jets (Andreopoulos and Rodi, 1984) at similar blowing ratios. The relatively uniform profiles found in the present study are probably due to the use of a short L/D delivery passage compared to the relatively long pipe inlets used in previous studies. Furthermore, the sharp entrance to the inlet tube (see Fig. 1) is expected to cause a separation region on the downstream side of the tube, which would become more severe for higher tube velocities. Turbulence intensity measurements, to be discussed in the next section, confirmed this expectation. Consequently, the increased separation region within the delivery passage at higher blowing ratios would account for the skewing of the jet inlet profiles toward the leading edge of the hole for the large blowing ratio. Since the hole geometry and passage length used in this study have significant effects on the exit profile of the jet, it is important to reiterate that the present configuration is more representative of inlets actually used for film-cooled turbine blades.

Downstream of the hole a "wake region" was formed by the blockage of the jet for all three blowing ratios. In each case a strong shear layer was formed between the low speed fluid in the wake and the lower edge of the jet. For $M = 0.25$, a shear layer also developed between the mainstream and the relatively low-speed fluid at the top edge of the jet. These shear layers merged to form a continuous shear region across the jet for this lowest blowing ratio. The velocity in the wake region was found to increase substantially with increasing blowing ratio indicating a greater ingestion of mainstream fluid below the jet for higher blowing ratios. A lateral influx of mainstream fluid below the jet was also indicated for $M = 0.5$ by a negative velocity gradient near the wall evident in Fig. 3 for $x/D > 3$. This can only be accounted for by an influx of high-speed fluid into the lower part of the wake. A similar effect was seen for $M = 1.0$, but not for $M = 0.25$.

Turbulence Quantities, Near Field. Measurements of the two components of the fluctutating velocity, u_{rms} and v_{rms}, are presented in terms of a turbulence level (TL) defined as

$$TL = (u_{rms}^2 + v_{rms}^2)^{1/2}/U_\infty$$

where U_∞ is the free-stream velocity. Contours of TL were obtained by interpolating the data, and the results for the three blowing ratios are shown in Figs. 5, 6, and 7. In each case several distinct regions of high turbulence were identified at the exit of the hole and immediately downstream. These regions are most apparent at the largest blowing ratio, $M = 1.0$ (Fig. 7). The smallest region of high turbulence is at the upstream edge of the jet as it exists from the hole. The turbulence here is likely due to the remnants of the boundary layer that develops within the hole along the upstream side.

For blowing ratios of $M = 0.5$ and $M = 1.0$, larger regions of high turbulence were found above the downstream side of the hole. The larger extent of these regions of turbulence suggests the existence of a much fuller turbulent boundary layer along the downstream surface of the hole. These turbulence data represent additional evidence that a separation region exists on the downstream side of the hole, following the sharp entrance. Furthermore, this would account for the significantly

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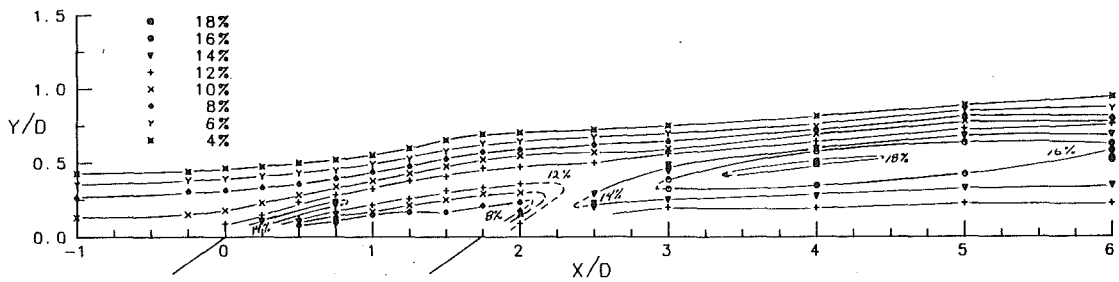

Fig. 5 Turbulence level, $(u^2_{rms} + v^2_{rms})^{1/2}/U_\infty$, contours for the near-hole region of the jet exit and $M = 0.25$

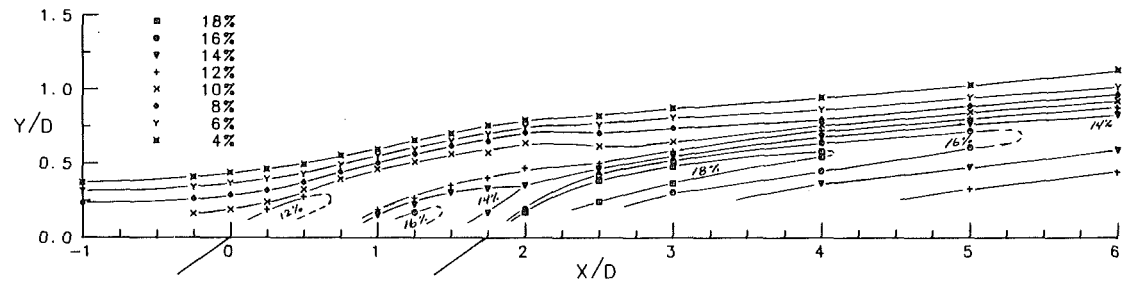

Fig. 6 Turbulence level, $(u^2_{rms} + v^2_{rms})^{1/2}/U_\infty$, contours for the near-hole region of the jet exit and $M = 0.5$

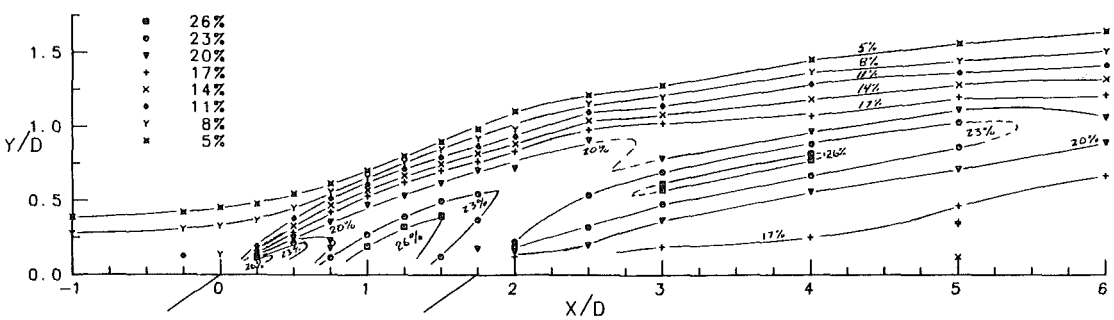

Fig. 7 Turbulence level, $(u^2_{rms} + v^2_{rms})^{1/2}/U_\infty$, contours for the near-hole region of the jet exit and $M = 1.0$

thicker boundary layer on this side of the hole. As discussed earlier, the size of this separated region is expected to decrease for decreasing blowing ratios, which would explain the decrease in the size of the turbulent region for $M = 0.5$ and the disappearance of this high turbulence region for $M = 0.25$.

In general, the highest level of turbulence was found along the bottom of the jet interface. Evidently this high turbulence is due to the shear layer that develops between the jet and the low-velocity wake below the jet. There was only a slight increase in the turbulence level in this region when the blowing ratio was increased from $M = 0.25$ to $M = 0.5$, but a dramatic increase from 18 percent to more than 26 percent occurred when the blowing ratio was increased to $M = 1.0$.

Comparing the above results to measurements by Yoshida and Goldstein (1984), at a location one diameter downstream of the hole for $M = 0.5$, shows there is a good agreement in the location of a maximum turbulence level at $y/D = 0.35$ in the shear layer below the jet. However, the maximum turbulence level of 12 percent was somewhat less than the 18 percent measured in this study. There are significant differences in the present results and those obtained by Jubran and Brown (1985) at a distance approximately two diameters downstream of the hole. As indicated in Fig. 7, for $M = 1.0$ and $x/D = 3.76$ the present measurements show a distinct peak in TL at $y/D = 0.7$, as opposed to maximum in the near-wall region ($y/D = 0.1$) measured by Jubran and Brown. A distinct peak in TL was also found in the present study for $M = 0.25$, but none was found by Jubran and Brown for $M = 0.2$.

Contours of the uv shear stress near the jet exit for the two lower blowing ratios, shown in Figs. 8 and 9, are very similar to the corresponding contours of TL shown previously. A small region of high uv at the upstream edge of the jet was found for both blowing ratios and is probably the remnants of the boundary layer that developed along the top side of the hole. At $M = 0.5$ a region of high turbulent shear stress exists above the downstream part of the jet exit, whereas at the lower blowing ratio of $M = 0.25$, the turbulent shear stress decreases at the jet exit. This is consistent with the contours of TL, and also indicates the existence of a separated region on the downstream side of the inlet tube, which is considerably reduced in size for the lower blowing ratio.

Maximum levels of uv shear stress were comparable for both blowing ratios and were found downstream of the jet at the interface between the jet fluid and the low velocitiy wake beneath the jet. Normalized with free-stream velocity, the peak value of $uv/U^2_\infty = -0.007$ is considerably larger than for a standard boundary layer flow, but consistent with the magnitude found in flows following separated regions such as the flow behind a backward-facing step (e.g., Chandrsuda and Bradshaw, 1981).

Downstream Development. The development of mean velocity profiles downstream of the jets was similar for all three

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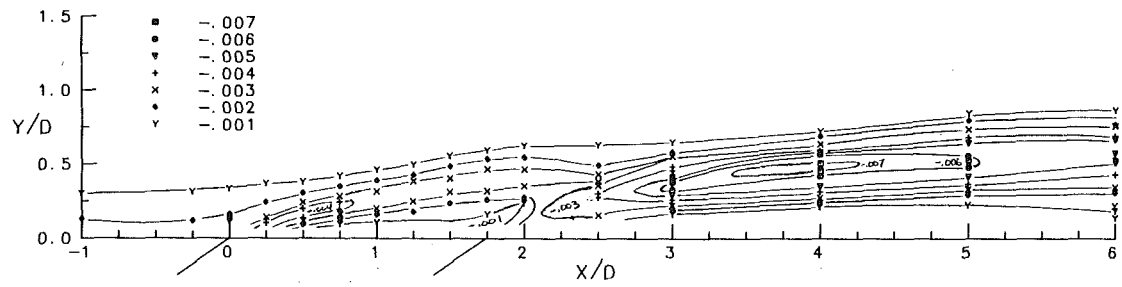

Fig. 8 Normalized shear stress, uv/V_∞^2, contours for the near-hole region of the jet exit and M = 0.25

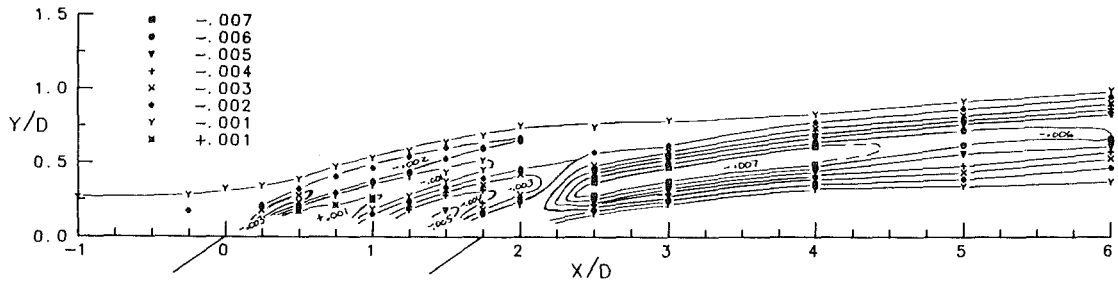

Fig. 9 Normalized shear stress, uv/V_∞^2, contours for the near-hole region of the jet exit and M = 0.5

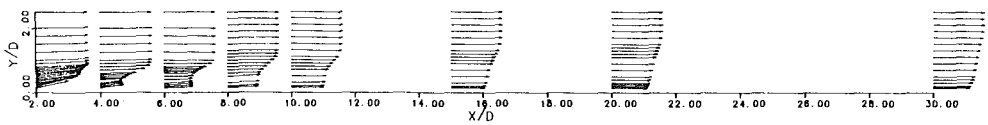

Fig. 10 Downstream development of the normalized mean velocity vectors along the jet centerline for M = 0.5

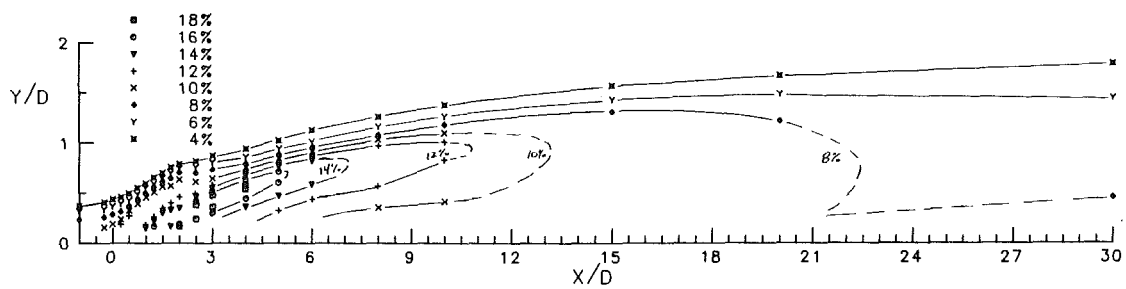

Fig. 11 Downstream development of the turbulence level, $(u_{rms}^2 + v_{rms}^2)^{1/2}/U_\infty$, for M = 0.5

blowing ratios, and representative results for $M = 0.5$ are shown in Fig. 10. These profiles show the shear layer that develops at the lower part of the jet is still apparent at $x/D = 10$, but has merged into the bottom wall boundary layer by $x/D = 30$. As shown in Table 1, boundary layer thickness measurements at $x/D = 30$ increased monotonically with increase in blowing ratio. However, the momentum thickness for $M = 0.5$ was substantially greater than for the higher and lower blowing ratios.

Contours of turbulence levels and uv shear stress downstream of the jet exit for $M = 0.5$ are given in Figs. 11 and 12, respectively. Note that the apparent trajectory of the jet in these figures is misleading because the vertical scales have been expanded for better resolution. Both turbulence quantities were found to decay rapidly by $x/D = 10$. Similar rapid decays in turbulence were found at the higher and lower blowing ratios. Moreover, despite the significantly higher turbulence levels for $M = 1.0$, turbulence levels for all three blowing ratios were essentially the same for $x/D > 10$, although the vertical extent of this turbulence was greater for higher blowing ratios. At $x/D = 30$ the turbulence levels and uv shear stress monotonically decreased moving away from the wall indicating that the flow has relaxed to a boundary layer flow.

Table 1 Boundary layer thickness at x/D = 30

M	δ/D	θ/D
0.25	1.2	0.12
0.5	1.9	0.19
1.0	2.5	0.15

Correspondence Between Turbulence Parameters and Mean Velocity Gradients. An important consideration from the standpoint of modeling is the degree of correspondence between turbulence parameters and mean velocity gradients. Using $M = 0.5$ as a representative blowing ratio, results for TL, uv, and dU/dy are presented for the region above and immediately behind the hole (Fig. 13), and for the development down-stream (Fig. 14). Each of the parameters in these figures has been normalized with the maximum measured value, which occurs at $x/D = 2.5$. Good agreement between the distribution of velocity gradients and the distribution of uv shear stress is immediately obvious from these two figures. The large positive shear generated at the top edge of the jet is evident from the profiles between $x/D = 0.5$ and $x/D = 2.0$, and the correspondence with uv was found to be excellent for this shear layer. Moreover, negative velocity gradients, which formed

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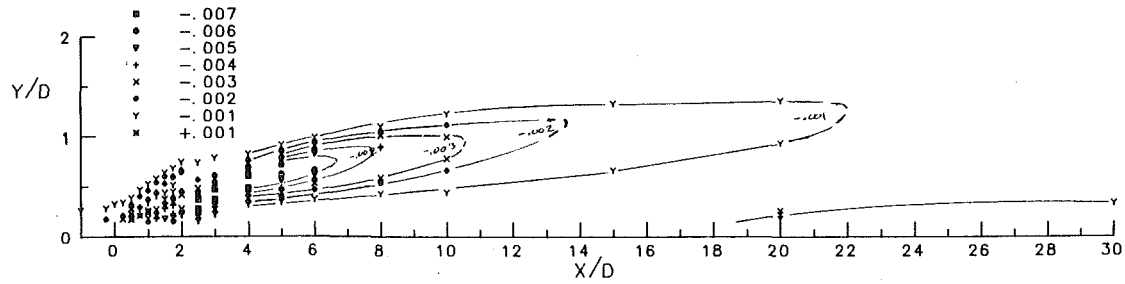

Fig. 12 Downstream development of the normalized shear stress, uv/U_∞^2, for $M = 0.5$

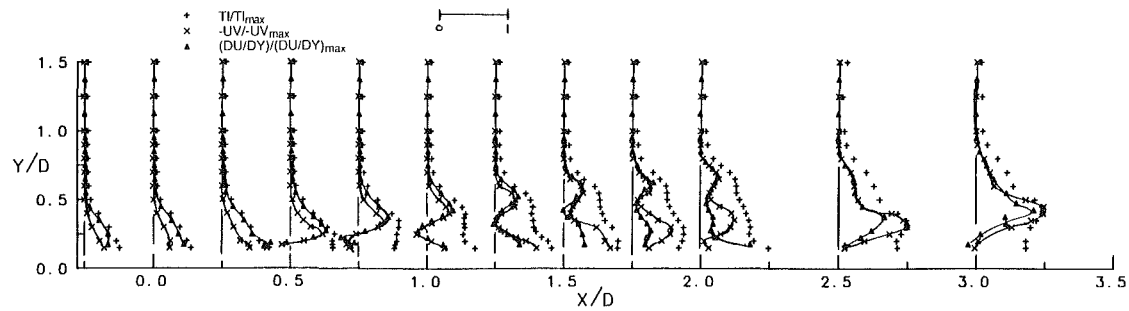

Fig. 13 Correspondence between the normalized turbulence quantities and the normalized mean velocity gradient for the near-hole region of the jet exit and $M = 0.5$

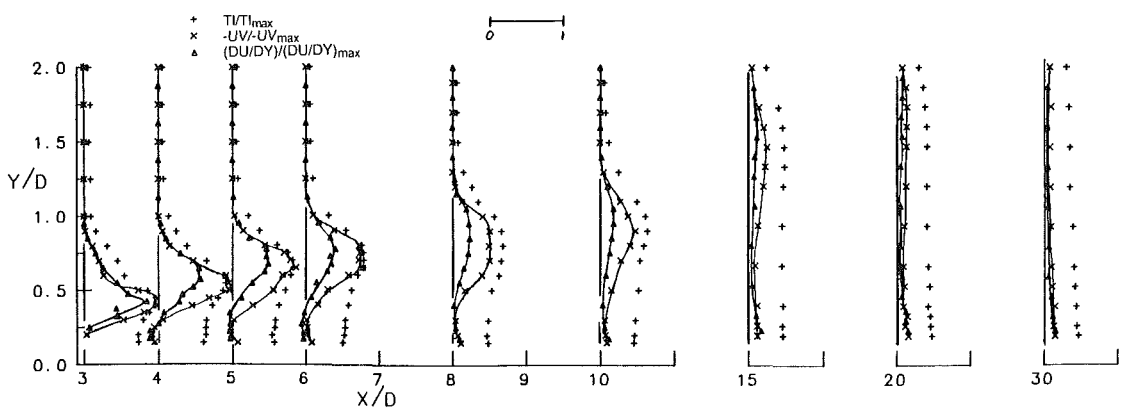

Fig. 14 Downstream development of the correspondence between the normalized turbulence quantities and the normalized mean velocity gradient for $M = 0.5$

below the top edge of the jet for $x/D = 0.5$ and 0.75, were found to correspond with peaks in positive uv shear stress. Note that the turbulence level remains high in this region where uv has a strong variation from negative to positive.

Immediately downstream of the back edge of the hole, at $x/D = 1.75$ and 2.0, a second uv peak appeared below the previous peak. Based on the contours of Fig. 9, it is apparent that this second peak is due to the remnants of the separated region within the inlet tube. These peaks in uv were associated with only mild peaks in velocity gradient.

At $x/D = 2.5$, the shear layer at the bottom of the jet dominates the flow. Good correspondence was found between the large velocity gradient and both the turbulence level and uv shear stress. As shown in Fig. 14, this good correspondence continues downstream although the magnitude of the velocity gradient decays faster than the uv shear stress. Beyond $x/D = 10$ the shear layer is no longer distinguishable.

Conclusions

The presence of a row of inclined jets issuing into a crossflow at a low blowing ratio ($M < 1$) causes a major disturbance to the hydrodynamic development of the turbulent boundary layer on the wall. All turbulence quantities are substantially enhanced in the region downstream of the jet. At ten diameters downstream, the presence of the jet is still clearly distinguishable in the profiles of the mean and fluctuating velocity components, but the flow is approaching normal wall boundary layer characteristics. The coincident occurrences of peaks in the mean velocity gradient and peaks in the turbulence quantities indicate that an eddy-viscosity model for the turbulence quantities may be applicable.

Computer simulation of jets-in-crossflow require that the flow field at the exit of the jet be specified. Previous studies indicate that a highly nonuniform velocity profile exists at the exit to jets issuing from long tubes. Measurements presented

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here, for the more realistic geometry of jets issuing from short tubes, indicate that a uniform velocity profile may be a better representation of actual film-cooling conditions.

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References

Andreopoulos, J., and Rodi, W., 1984, "Experimental Investigation of Jets in a Crossflow," *Journal of Fluid Mechanics*, Vol. 138, pp. 93–127.

Crabb, D., Durao, D. F. G., and Whitelaw, J. H., 1981, "A Round Jet Normal to a Crossflow," ASME *Journal of Fluids Engineering*, Vol. 103, pp. 142–152.

Chandrsuda, C., and Bradshaw, P., 1981, "Turbulence Structure of a Reattaching Mixing Layer," *Journal of Fluid Mechanics*, Vol. 110, pp. 171–194.

Jubran, B., and Brown, A., 1985, "Film Cooling From Two Rows of Holes Inclined in the Streamwise and Spanwise Directions," ASME *Journal of Engineering for Gas Turbines and Power*, Vol. 107, pp. 84–90.

Kadotani, K., and Goldstein, R. J., 1979,, "On the Nature of Jets Entering a Turbulent Flow, Part A. Jet-Mainstream Interaction," ASME *Journal of Engineering for Power*, Vol. 101, pp. 459–465.

Yoshida, T., and Goldstein, R. J., 1984, "On the Nature of Jets Issuing From a Row of Holes Into a Low Reynolds Number Mainstream Flow," ASME *Journal of Engineering for Gas Turbines and Power*, Vol. 106, pp. 612–618.

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